Helios HX-48 Commissioning Manual

HXM-48-2026 - bilingual procedural document

Purpose

This manual defines commissioning and maintenance steps for the Helios HX-48 grid-connected inverter. It is intended for trained technicians working under an approved isolation permit. The equipment family identifier is HX-48; HX-84 appears in a training appendix and is not compatible with the procedures in this edition.

Document control

Edition 1.3 was approved on 22 September 2026. Field annotations do not change the controlled procedure until engineering publishes a new edition. Screenshots may assist navigation, but settings must be verified against the displayed firmware version and the approved parameter table.

Safety boundary

This synthetic manual does not authorize real electrical work. It contains no site access codes, customer credentials or emergency phone numbers. Test questions must be answered only from the authored fixture and must not be treated as operational advice.

Reading conventions

WARNING denotes a potential safety consequence, CAUTION denotes equipment damage and NOTE records useful context. A numbered step is mandatory only within its procedure. Similar values in troubleshooting examples are deliberately retained as retrieval negatives.

1. Safe shutdown sequence

Numbered procedure with time-dependent evidence

Preparation

Record the active power, DC voltage and firmware build before changing state. Open the approved work order and confirm that the serial number on the enclosure matches the asset record. A screen label alone is insufficient because replacement control boards may retain an earlier site name.

Isolation

Set the operating mode to STANDBY, open the AC isolator and then open the DC disconnect. After both sources are open, wait 7 minutes before removing the service cover. The nearby 70-second delay applies only to a communications restart and must not be used for electrical isolation.

Verification

Measure the specified test points with an approved meter and confirm the indicator is dark. If voltage remains above the documented limit, replace the cover, secure the work area and escalate through the site procedure. Do not shorten the waiting period because a display has turned off.

Return to service

Reinstall all barriers, close the DC disconnect, close the AC isolator and leave STANDBY only after the self-test completes. Record the self-test identifier and final power value in the work order.

2. Mechanical and electrical settings

Native table, units and near-value distractors

Fastener control

Use a calibrated tool and record the final value. The DC terminal torque is 6.2 N m. A legacy HX-36 bulletin specifies 5.8 N m, but that value is not approved for HX-48 equipment.

Connection	Conductor	Torque	Inspection	Reference
DC positive	35 mm2	6.2 N m	each service	HX48-T01
DC negative	35 mm2	6.2 N m	each service	HX48-T02
AC phase	50 mm2	8.4 N m	annual	HX48-T03
Protective earth	25 mm2	7.1 N m	annual	HX48-T04
Signal shield	1.5 mm2	0.8 N m	installation	HX48-T05

Limits

The maximum continuous AC current is 69 A at the nominal test condition. A commissioning screen may briefly display 76 A during its simulated check; that screen value is not the continuous rating.

3. Control path overview

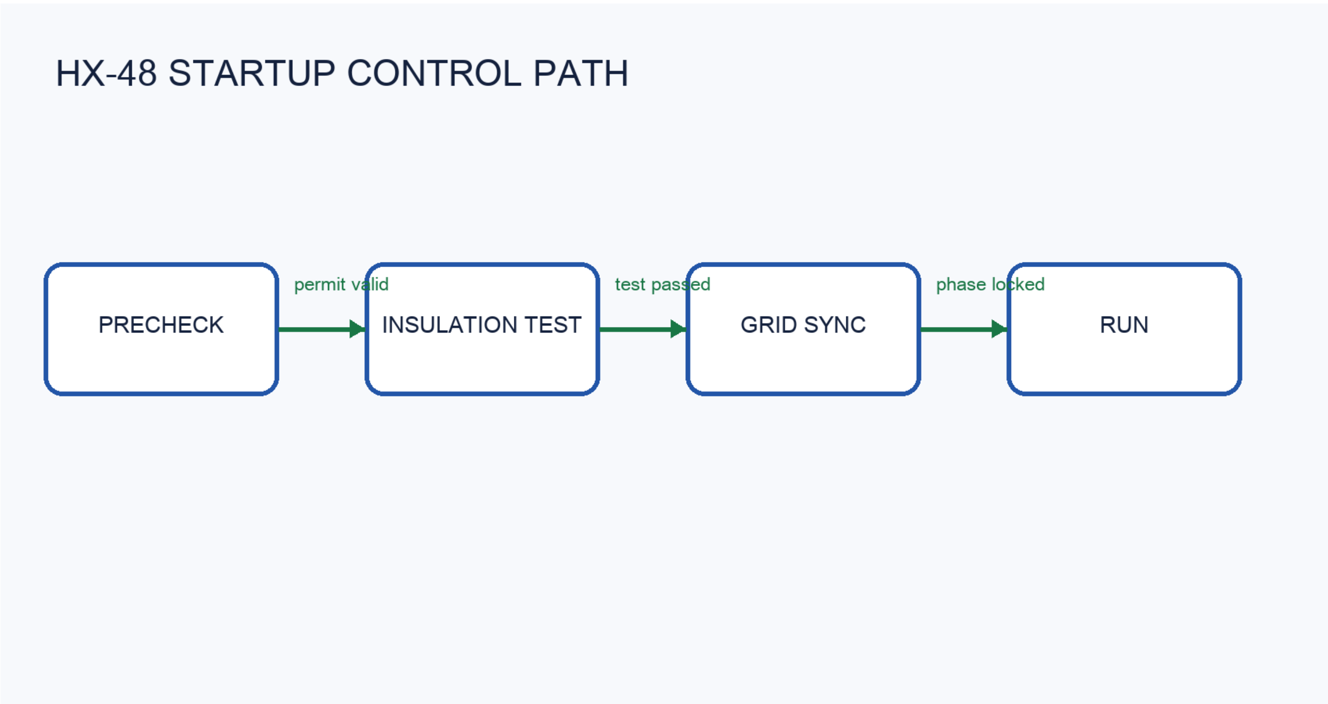
Raster-only process diagram

Diagram scope

Figure 3 shows the control path used during startup. Node names and arrow destinations are present only in the raster diagram. The surrounding paragraph intentionally avoids restating the route so visual evaluation cannot be bypassed through the text layer.

Failure handling

A failed insulation check stops the sequence and preserves the event record. The technician must identify the failed stage before restarting; repeated start commands are not a substitute for diagnosis.



4. 固件与告警处理

中文说明、版本约束与连续条件

固件范围

本版手册批准的最低固件版本为 4.7.2。版本 4.7.1 可以读取历史记录，但不能启用新的绝缘测试流程。升级完成后必须重新读取设备身份信息，确认型号仍为 HX-48。

温度告警

散热器温度连续三个采样周期达到或超过 82 摄氏度时进入人工复核。单次 82 摄氏度记录只产生观察事件。培训屏幕中的 28 摄氏度是界面演示值，不是告警阈值。

复位条件

只有在温度回落、风扇自检通过且工单记录完整后才能复位。远程操作员可以发起诊断，现场技术人员负责确认物理通风路径。两种角色不能互相替代。

记录保留

固件包校验值、参数导出和升级结果至少保存 24 个月。截图缺少结构化参数时只能作为辅助证据，不能单独证明设置正确。

5. Troubleshooting and exclusions

Symptoms, misleading matches and no-answer controls

No grid sync

Confirm phase order and grid window before replacing hardware. A communication timeout can appear nearby in the log but does not prove a phase error.

Repeated fan alarm

Inspect airflow and connector seating. Do not change the 82 degree review rule to suppress alerts.

Rejected draft

An engineering draft proposed a 5-minute cover wait. Safety review rejected it; the approved shutdown sequence remains 7 minutes.

Unsupported facts

The manual contains no installer password, private service number, site address or electricity tariff. Those questions require abstention.

Citation practice

Procedural answers cite the numbered sequence, parameter answers cite the table row and visual route answers cite the diagram region.